SOFTWARE REQUIREMENTS SPECIFICATION

COLLEGE ADMISSION MANAGEMENT SYSTEM

A proposed web-based system for organized and transparent admission processing

STUDENT | **APPLICATION** | **ADMISSION**

SUBMISSION DETAILS	TO BE COMPLETED BY STUDENT
Student Name	_____
Register Number	_____
Department	_____
College Name	_____
Faculty Name	_____
Submission Date	_____

Prepared for first-year engineering coursework

DOCUMENT CONTROL

Item	Details
Document Title	Software Requirements Specification (SRS)
Project	College Admission Management System
Document Version	1.0
Document Status	Proposed specification; software is not yet implemented
Prepared By	[Student Name]
Review Status	For academic review

REVISION HISTORY

Version	Date	Description	Author
1.0	[Submission Date]	Initial SRS draft for the proposed system	[Student Name]

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Note: In Microsoft Word, the table of contents can be replaced with an automatic TOC after applying Word heading styles. Page numbers may change when the document is edited.

1. INTRODUCTION

1.1 Purpose of this document

This Software Requirements Specification (SRS) defines the proposed features, users, interfaces, data, constraints, and quality expectations for a College Admission Management System. It gives students, admission staff, administrators, and developers a shared description of what the system is expected to do. It follows an IEEE-style structure adapted for a first-year engineering assignment.

1.2 Project overview

A College Admission Management System is a proposed software application that helps a college receive, organize, review, and track student admission applications. Students can create an account, enter academic details, upload required documents, submit an application, and check its status. Authorized admission officers can review applications, verify eligibility, request clarification, and record a decision. Administrators manage system users and configuration.

1.3 Need for the system

In a manual process, forms and documents may be collected on paper or through separate channels. Staff may need to enter the same information more than once, search through files, contact applicants individually, and prepare reports manually. These activities can take time and may lead to missing documents, duplicate records, data-entry errors, or unclear status updates.

The proposed system centralizes application information in a controlled database. It is intended to make the process easier to follow, improve record consistency, support timely review, and give applicants a clearer view of progress. It supports admission work but does not replace official admission rules or the final authority of the college.

1.4 Definitions and abbreviations

Term	Meaning
SRS	Software Requirements Specification
FR	Functional Requirement: a service or behaviour the system must provide
NFR	Non-Functional Requirement: a quality or constraint of the system
UI	User Interface
DBMS	Database Management System
Applicant	A student who creates or submits an admission application
Eligibility check	A comparison of entered details with configured admission criteria

2. PURPOSE AND OBJECTIVES

The purpose is to specify a consistent, secure, and easy-to-use digital workflow for college admission applications.

- Reduce paper-based records and repeated manual data entry.
- Allow students to complete an application through a single online interface.
- Improve accuracy by validating required fields and keeping related records together.
- Save time when searching, reviewing, and summarizing applications.

- Help students track whether an application is submitted, under review, awaiting information, approved, or rejected.
- Support admission officers with a clear review queue and recorded decisions.
- Provide administrators with controlled user management and summary reports.

3. PROJECT SCOPE

3.1 In scope

- Student account registration, login, and profile maintenance.
- Online application form and academic-detail entry.
- Upload and storage of specified supporting documents.
- Basic eligibility checks using criteria configured by authorized staff.
- Application review, remarks, and decision recording by admission officers.
- Student-facing application status and basic notifications within the system.
- Administrative management of users, departments, admission settings, and reports.

3.2 Out of scope

- Making official admission policies or changing government/college rules.
- Guaranteeing admission, seat availability, or a particular outcome.
- Online fee payment, scholarships, hostel allocation, transport, and counselling allotment in the initial version.
- Automatic authentication of documents with external government databases.
- Native Android/iOS applications; the initial proposal is a responsive web application.
- Integration with external college ERP, payment gateways, or email/SMS providers in the initial version.

These items may be considered as future enhancements if the college approves them.

4. OVERALL SYSTEM DESCRIPTION

4.1 Product perspective

The system is proposed as a web application with a browser-based interface, an application server that applies rules and permissions, and a database that stores users, applications, departments, documents, and review history.

4.2 Major system functions

- Manage user accounts and authenticated sessions.
- Collect student profile and academic information.
- Accept an application and supporting files.
- Check whether mandatory information is present and compare data with configured eligibility criteria.
- Route applications to authorized staff for review.
- Record a decision and show the current status to the student.
- Generate summary reports for authorized administrators.

4.3 Operating environment

Component	Proposed environment
Client device	Desktop, laptop, tablet, or smartphone with a modern web

	browser
Operating system	Windows, Linux, macOS, Android, or iOS supported by the browser
Application layer	Web server/application service; implementation technology to be selected
Database	Relational DBMS such as MySQL or PostgreSQL; final choice is a design decision
Network	Internet or college network connection using HTTPS in a deployed environment

4.4 Users and characteristics

Students use the system to submit and track their own applications. Admission officers are expected to understand the college's admission procedure and review records carefully. Administrators are responsible for user access, configuration, and reports. Access should be role-based so users see only the functions needed for their work.

4.5 Assumptions and dependencies

- The college supplies approved admission criteria, required document lists, and status definitions.
- Students have access to a supported browser and a usable internet connection.
- Authorized staff maintain accurate department and admission-cycle information.
- Document storage, database hosting, backups, and security configuration are available before deployment.
- Any email/SMS service, if added later, depends on a valid provider and configuration.

4.6 General limitations

The system depends on correct information entered by applicants and on the accuracy of configured admission rules. It cannot independently guarantee that uploaded documents are genuine. Service availability depends on hosting, network access, maintenance, and backup arrangements.

5. USER ROLES

Role	Responsibilities	Access boundaries
Student	Register and log in; maintain personal profile; complete application; enter academic details; upload documents; submit application; view status and requests for clarification.	Can access and update permitted information belonging to their own account; cannot review or decide other applications.
Admission Officer	View assigned/submitted applications; check completeness; compare details with approved criteria; request further information; add review remarks; record a recommendation or decision where authorized.	Can access application records required for admission duties; cannot manage system-wide security settings unless separately authorized.
Administrator	Create or disable staff accounts; assign roles; manage departments and admission-cycle settings; configure required fields and criteria based on approved policy; view system-level reports and audit records.	Administrative privileges should be limited to authorized personnel and important actions should be logged.

6. FUNCTIONAL REQUIREMENTS

Each requirement below describes an expected system behaviour. Requirements are proposed and have not been implemented or tested.

FR-01: Student Registration and Login

The system shall allow a student to register using required identity and contact fields, create login credentials, and sign in. It shall validate required fields and reject duplicate account identifiers. Passwords must not be stored as plain text.

FR-02: Student Profile Management

The system shall allow an authenticated student to view and update permitted profile fields such as contact information and address. Changes to critical identity fields may require staff review.

FR-03: Online Admission Application

The system shall provide an application form for the selected admission cycle and department/program. It shall allow the student to save a draft and submit the form after required fields are completed.

FR-04: Academic Details Entry

The system shall collect relevant academic information, such as qualification, board or institution, year of completion, and marks/grades, according to the approved form. It shall validate field formats and acceptable ranges.

FR-05: Document Upload

The system shall allow students to upload the specified supporting documents in permitted file formats and sizes. It shall show upload success or a clear error and associate each file with the correct application.

FR-06: Eligibility Verification

The system shall compare entered academic details with eligibility criteria configured by authorized staff. It shall show a preliminary result such as 'Criteria appear met', 'Not met', or 'Requires manual verification'. The result shall not replace the official admission decision.

FR-07: Application Review

The system shall provide authorized admission officers with an application review view containing submitted details, documents, eligibility result, and review history. Officers shall be able to record remarks and request additional information or record an authorized decision.

FR-08: Admission Status Tracking

The system shall show each student the current status of their own application and, where appropriate, the latest action date and any required next step. Possible statuses include Draft, Submitted, Under Review, Further Information Required, Approved, and Rejected.

FR-09: Notifications

The system shall display in-system notifications when an application is submitted, its status changes, or additional information is requested. External email/SMS notifications are not required in the initial version.

FR-10: Admission Reports

The system shall allow authorized administrators to view or export summary reports, such as total applications by department and counts by status. Reports shall respect user permissions and avoid exposing unnecessary personal data.

6.1 Business rules

- An application cannot be submitted until mandatory fields and required documents are present.
- A student can view only applications linked to their own account.
- Only authorized staff can change review status or record an admission decision.
- Eligibility results are preliminary unless the college formally defines them as part of an approved workflow.
- Every submitted application should have a unique application identifier.
- Important status changes should record the acting user and timestamp.

7. NON-FUNCTIONAL REQUIREMENTS

ID / Quality	Requirement
NFR-01 Security	The system shall authenticate users and enforce role-based access control. Passwords shall be stored using a suitable salted password-hashing method. Sensitive traffic shall use HTTPS in deployment.
NFR-02 Performance	Under normal expected college usage, common pages should respond within approximately 3 seconds for 95% of requests, excluding slow network connections and large file transfers. This target must be validated against the final hosting environment.
NFR-03 Usability	Forms shall use clear labels, instructions, field validation, and understandable error messages. The interface should work on common desktop and mobile screen sizes.
NFR-04 Reliability	The system should handle invalid input without crashing and should provide a clear message when an operation fails. Database backups and a recovery procedure should be defined before deployment.
NFR-05 Maintainability	The application should be divided into understandable modules, use consistent naming, and include basic technical documentation so future developers can update it.
NFR-06 Compatibility	The web interface should support current stable versions of commonly used browsers, subject to testing before release.
NFR-07 Data protection	Personal data and documents shall be accessible only to authorized roles. Data collection should be limited to admission needs, retention should follow college policy, and records should be securely removed when permitted.
NFR-08 Auditability	Changes to application status, eligibility settings, and user roles should record the user, time, and action where practical.
NFR-09 Accessibility	The interface should use readable contrast, keyboard-accessible controls, descriptive labels, and meaningful error messages.
NFR-10 Scalability	The design should allow capacity to be increased if application volume grows, subject to hosting and database design.

8. EXTERNAL INTERFACE REQUIREMENTS

8.1 User interface

The interface should provide a consistent navigation menu and separate dashboards for students, admission officers, and administrators. Forms should group related fields, identify mandatory items, display upload limits, and show clear confirmation after saving or submitting. Status labels should be consistent throughout the application.

8.2 Software interfaces

The proposed application will communicate with a relational database through a server-side data-access layer. If external email, SMS, payment, or college ERP services are introduced later, their APIs and authentication methods must be specified separately. No external integration is assumed for the initial version.

8.3 Database interface

The application server shall read and write structured records in the database using validated queries or a suitable data-access framework. Database credentials must not be exposed in the browser. Access permissions and backups must be configured by the deployment team.

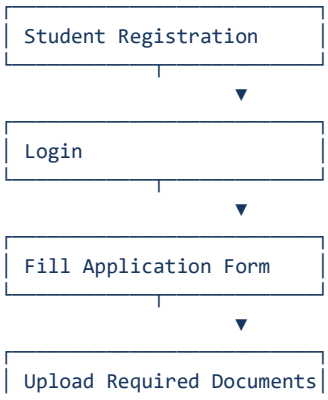
8.4 Basic hardware requirements

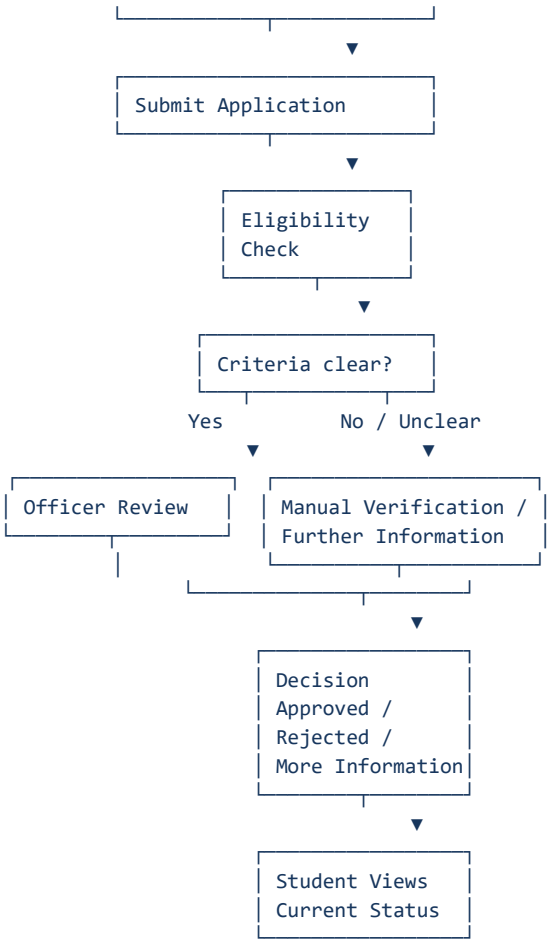
Component	Minimum planning requirement
Student device	A smartphone, tablet, laptop, or desktop capable of running a supported browser
Staff device	A computer with a supported browser and reliable network access
Server	A server or approved hosting service sized for the expected number of users and uploaded documents
Storage	Sufficient secure storage for database backups and uploaded files; capacity to be estimated from actual usage
Network	Reliable internet/college network; HTTPS support for deployed access

Exact CPU, memory, and storage specifications should be selected after the expected application volume and deployment environment are known.

9. SYSTEM WORKFLOW

The following flow describes the expected application journey. The eligibility check is preliminary; an admission officer reviews the application before an authorized decision is recorded.





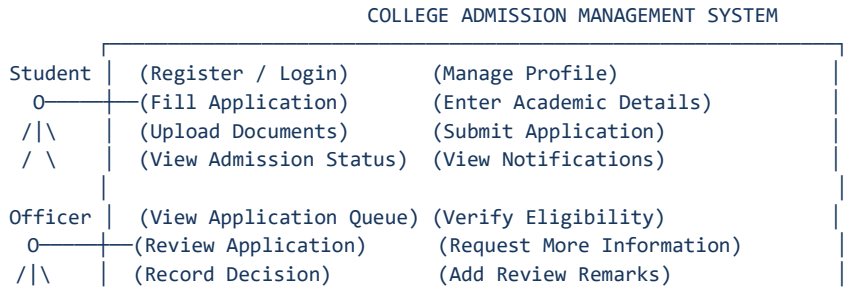
Decision handling: Approved and Rejected are final statuses only when recorded by an authorized officer. If information is missing or unclear, the application should be marked for further verification rather than automatically rejected.

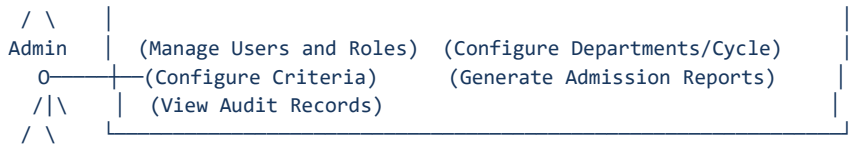
9.1 Workflow summary

1. Registration and login establish the student's account.
2. The student enters the application details and uploads required files.
3. The system validates completeness and performs a preliminary eligibility check.
4. An admission officer reviews the application and may request more information.
5. An authorized decision is recorded, and the student can view the updated status.

10. USE-CASE DIAGRAM

The diagram below shows the main interactions between the three user roles and the system. It is a simplified UML-style view for this academic specification.





Legend: Each actor is a role. The items in parentheses are use cases (goals the role can achieve using the system). The diagram is simplified for readability; a formal UML drawing may be produced in a diagramming tool if required by the faculty.

10.1 Use-case descriptions

Use case	Primary actor	Expected outcome
Submit application	Student	A complete application is stored with a unique application ID and Submitted status.
Upload documents	Student	Permitted files are attached to the student's application.
Verify eligibility	Admission Officer / System	A preliminary eligibility result is recorded for review.
Review application	Admission Officer	The officer records remarks or requests further information.
Record decision	Authorized Admission Officer	The application status is updated and the action is logged.
Manage users and criteria	Administrator	Authorized accounts, roles, and approved configuration are maintained.
Generate reports	Administrator	A summary of applications by department/status is available to authorized staff.

11. DATABASE DESIGN

The proposed design uses a relational database. The following entities are logical starting points; exact data types, keys, validation rules, and retention periods should be finalized during detailed design.

11.1 Main entities and attributes

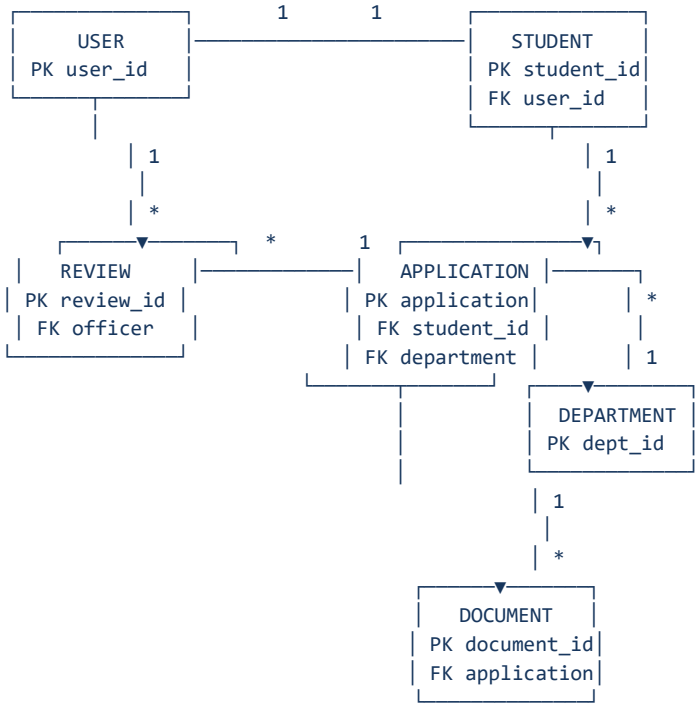
Entity	Important attributes	Purpose
User	user_id (PK), email_or_username, password_hash, role, account_status, created_at, last_login_at	Stores login identity, role, and account state.
Student	student_id (PK), user_id (FK), full_name, date_of_birth, phone, address, identity_reference_if_required	Stores student profile details. Collect only fields approved as necessary.
Department	department_id (PK), department_name, program_name, intake_cycle, active_status	Lists departments/programs accepting applications.
Application	application_id (PK), student_id (FK), department_id (FK), application_cycle, academic_summary, eligibility_result, status, submitted_at, updated_at	Stores one admission application and its current state.
Document	document_id (PK), application_id (FK), document_type, storage_reference, uploaded_at, verification_status, reviewer_note	Stores metadata and secure references to uploaded files; files need not be stored directly in a database row.
Review	review_id (PK), application_id (FK), officer_user_id (FK), action, remarks, action_at	Maintains a history of reviews, requests, and decisions.
Notification	notification_id (PK), user_id (FK),	Stores in-system notifications.

	application_id (FK), message, created_at, read_at	

11.2 Relationships

- One User account may be linked to one Student profile; staff accounts use the User role field.
- One Student can submit multiple Applications across departments or admission cycles, subject to college rules.
- One Department can be linked to many Applications.
- One Application can have multiple Documents and multiple Review records.
- One User may receive multiple Notifications.
- One admission officer may create multiple Review records.

11.3 Entity relationship diagram (logical)



Cardinality note: 1 means one record and * means many records. The diagram is a logical overview and omits some entities (such as Notification) to keep the relationships readable.

11.4 Data integrity and security

- Primary keys uniquely identify records; foreign keys link related records.
- Required fields and allowed status values should be validated by the application and database.
- Uploaded files should be stored in protected storage and referenced using a non-public identifier.
- Passwords must be hashed; access to personal information should be role-restricted.
- Database backups and a retention/deletion policy should be established before real student data is used.

12. ASSUMPTIONS AND CONSTRAINTS

12.1 Assumptions

- The college will provide current admission criteria and approve any changes.
- Students will provide truthful information and legible documents.
- Admission officers will review applications within the college's operational process.
- Administrators will manage roles carefully and deactivate accounts when access is no longer needed.

12.2 Constraints

- The initial version is limited to a browser-based application and basic admission workflow.
- The system's eligibility output is only as accurate as the configured criteria and entered data.
- Availability depends on hosting, connectivity, maintenance, and recovery arrangements.
- File type and file-size limits must be set according to storage and security requirements.
- External integrations are excluded until access, API specifications, and approvals are available.

12.3 Privacy and policy requirements

The system should collect only information required for admission, explain why information is needed, restrict access by role, protect files in storage and transit, and follow applicable law and college policy. Sensitive documents should not be made available through guessable or public URLs. The college must define retention periods, access approval, correction procedures, and secure deletion rules.

13. TEST CASES

These test cases describe planned checks for the proposed requirements. Since this document specifies a system that has not been implemented, every status is marked 'Not Tested'.

Test Case ID	Feature	Test input / action	Expected result	Status
TC-01	Registration	Enter valid required details and a unique email/username.	Account is created and a confirmation is shown.	Not Tested
TC-02	Registration	Register using an already registered email/username.	System rejects duplicate identifier and explains the issue.	Not Tested
TC-03	Login	Enter valid credentials for an active account.	User is authenticated and sent to the correct role dashboard.	Not Tested
TC-04	Login	Enter an incorrect password.	Access is denied and a safe error message is displayed.	Not Tested
TC-05	Application submission	Complete all mandatory fields and submit the form.	Application is saved with a unique ID and Submitted status.	Not Tested
TC-06	Application submission	Attempt to submit with a mandatory field missing.	Submission is blocked and missing field(s) are identified.	Not Tested
TC-07	Document upload	Upload a permitted file type within the configured size limit.	File is accepted and linked to the correct application.	Not Tested
TC-08	Document upload	Upload a prohibited file type or oversized file.	Upload is rejected with a clear explanation.	Not Tested
TC-09	Eligibility check	Enter academic details that satisfy configured	System records a preliminary 'Criteria	Not Tested

		criteria.	appear met' result.	
TC-10	Eligibility check	Enter details that are incomplete or outside configured criteria.	System flags the case for manual verification or indicates criteria not met as configured.	Not Tested
TC-11	Status tracking	Student opens the status page for their own application.	Current status and relevant next step are displayed.	Not Tested
TC-12	Access control	Student attempts to open another student's application.	System denies access and does not reveal private data.	Not Tested
TC-13	Officer review	Authorized officer requests additional information.	Status changes to Further Information Required and action is logged.	Not Tested
TC-14	Decision recording	Authorized officer records an approved or rejected decision.	Status is updated and the student can view it.	Not Tested
TC-15	Reports	Administrator generates an application count by status.	Report totals match the records included in the selected filter.	Not Tested

13.1 Acceptance criteria

- All mandatory functional requirements have corresponding verification checks.
- Role permissions prevent students from accessing other students' records.
- Application submission validates required fields and document requirements.
- Status changes are visible to the appropriate user and recorded in review history.
- Reports use the same status definitions as the application workflow.

14. FUTURE ENHANCEMENTS

- Secure online application-fee payment with transaction confirmation and reconciliation.
- Email and SMS alerts for submission, missing documents, and decision updates.
- Integration with the college ERP, student information system, or approved counselling platform.
- Digital document verification and duplicate-document detection, subject to policy and reliable services.
- Appointment scheduling for document verification or counselling.
- Analytics dashboards showing application trends by department and admission cycle.
- Accessibility improvements and additional language options based on student needs.
- Mobile application or progressive web app if usage needs justify it.

15. CONCLUSION

The proposed College Admission Management System is intended to provide a structured digital process for student registration, application submission, document handling, eligibility checking, staff review, and status tracking. By centralizing information and applying consistent validation and access controls, it is expected to reduce paperwork, improve record organization, save staff time, and make progress easier for students to understand.

This SRS defines requirements and design proposals only. It does not claim that the software has been developed, deployed, or tested. The next step would be to review these requirements with the college or faculty, confirm admission rules, prepare detailed screen and database designs, and then implement and test the approved system.

APPENDIX A: GLOSSARY

Term	Definition
Admission cycle	A defined period during which applications are accepted and processed.
Application ID	A unique reference assigned to an admission application.
Audit record	A record of an important action, including who performed it and when.
Eligibility	Whether entered details appear to satisfy specified admission criteria.
Role-based access control	A method that gives users permissions according to their assigned role.
Status	The current stage of an application in the admission workflow.

APPENDIX B: ACADEMIC SUBMISSION CHECKLIST

- Replace all cover-page placeholders with the correct student and college details.
- Confirm that the department and faculty name are written as required by the college.
- Review the admission workflow and criteria with the faculty before submission.
- Update the table of contents and page numbers in Word after making edits.
- Export a PDF from Microsoft Word and verify that tables and diagrams remain readable.